Sherry Yang

Contents

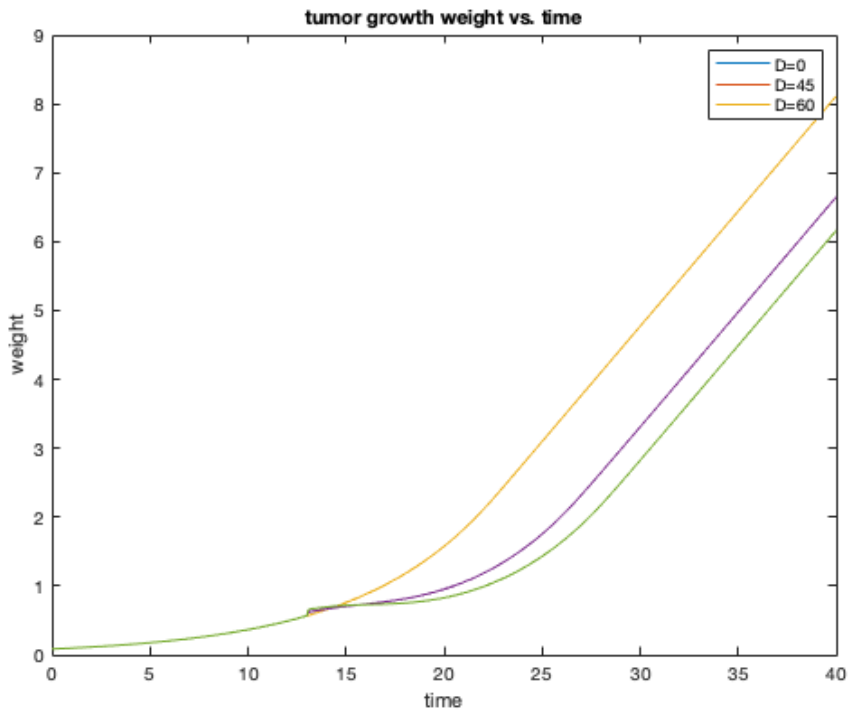
- hw3
- Oct 27th, 2023

hw3

Oct 27th, 2023

```
D1 = 0 ;
D2 = 45 ;
D3 = 60 ;
t = linspace (0,40,1000);

w_t_0 = RungeKutta(D1);
w_t_45 = RungeKutta(D2);
w_t_60 = RungeKutta(D3);

figure(1);
plot(t,w_t_0)
hold on
plot(t,w_t_45)
plot(t,w_t_60)
xlabel('time')
ylabel('weight')
title('tumor growth weight vs. time')
legend('D=0','D=45','D=60')

% tumor growth slow down up until 13 days, where it almost plateaued and
% then it picked up again and grew exponentially


function w = RungeKutta(D)
    t = linspace(0,40,1000);
    lambda0= 0.146 ;
    lambda1=0.334;
    w0 = 0.085;
    V = 4.85;
    phi = 20 ;
    k1 = 0.469 ;
    k2 = 0.842;
    k10 = 13.272 ;
    k12 = 0.276;
    k21 = 1.4784;
    dt = t(2)-t(1);
    x0 = [w0; 0; 0; 0];
    t0 = 13;

    x1 = zeros(1, length(t));
    x2 = zeros(1, length(t));
    x3 = zeros(1, length(t));
    x4 = zeros(1, length(t));
    x1(1) = x0(1);
    x2(1) = x0(2);
    x3(1) = x0(3);
    x4(1) = x0(4);

    alpha = 13.582;
    beta = 1.445;
```

```
    w = zeros(1, length(t));
    c = zeros(1, length(t));


    dx1_dt = @(c, w, x1, time) ((lambda0 * x1) / ((1 + ((lambda0 / lambda1) * w)^phi)^(1/phi))) - k2 * c * x1;
    dx2_dt = @(c, x1, x2, time) k2 * c * x1 - k1 * x2;
    dx3_dt = @(x2, x3, time) k1 * (x2 - x3);
    dx4_dt = @(x3, x4, time) k1 * (x3 - x4);

    A = (D / V) * ((alpha - k21) / (alpha - beta));
    B = -(D / V) * ((beta - k21) / (alpha - beta));

    for i = 1:length(t)
        w(i) = x1(i) + x2(i) + x3(i) + x4(i);

        if t(i) <= 13
            c(i) = 0;
        else
            c(i) = A * exp(-alpha * (t(i) - t0)) + B * exp(-beta * (t(i) - t0));
        end

        dx1_k1 = dx1_dt(c(i), w(i), x1(i), t(i));
        dx1_k2 = dx1_dt(c(i), w(i), x1(i) + dx1_k1 * dt / 2, t(i) + dt / 2);
        dx1_k3 = dx1_dt(c(i), w(i), x1(i) + dx1_k2 * dt / 2, t(i) + dt / 2);
        dx1_k4 = dx1_dt(c(i), w(i), x1(i) + dx1_k3 * dt, t(i) + dt);

        dx2_k1 = dx2_dt(c(i), x1(i), x2(i), t(i));
        dx2_k2 = dx2_dt(c(i), x1(i) + dx2_k1 * dt / 2, x2(i) + dx2_k1 * dt / 2, t(i) + dt / 2);
        dx2_k3 = dx2_dt(c(i), x1(i) + dx2_k2 * dt / 2, x2(i) + dx2_k2 * dt / 2, t(i) + dt / 2);
        dx2_k4 = dx2_dt(c(i), x1(i) + dx2_k3 * dt, x2(i) + dx2_k3 * dt, t(i) + dt);

        dx3_k1 = dx3_dt(x2(i), x3(i), t(i));
        dx3_k2 = dx3_dt(x2(i) + dx3_k1 * dt / 2, x3(i) + dx3_k1 * dt / 2, t(i) + dt / 2);
        dx3_k3 = dx3_dt(x2(i) + dx3_k2 * dt / 2, x3(i) + dx3_k2 * dt / 2, t(i) + dt / 2);
        dx3_k4 = dx3_dt(x2(i) + dx3_k3 * dt, x3(i) + dx3_k3 * dt, t(i) + dt);

        dx4_k1 = dx4_dt(x3(i), x4(i), t(i));
        dx4_k2 = dx4_dt(x3(i) + dx4_k1 * dt / 2, x4(i) + dx4_k1 * dt / 2, t(i) + dt / 2);
        dx4_k3 = dx4_dt(x3(i) + dx4_k2 * dt / 2, x4(i) + dx4_k2 * dt / 2, t(i) + dt / 2);
        dx4_k4 = dx4_dt(x3(i) + dx4_k3 * dt, x4(i) + dx4_k3 * dt, t(i) + dt);

        x1(i + 1) = x1(i) + (dx1_k1 + 2 * dx1_k2 + 2 * dx1_k3 + dx1_k4) * dt * (1/6);
        x2(i + 1) = x2(i) + (dx2_k1 + 2 * dx2_k2 + 2 * dx2_k3 + dx2_k4) * dt * (1/6);
        x3(i + 1) = x3(i) + (dx3_k1 + 2 * dx3_k2 + 2 * dx3_k3 + dx3_k4) * dt * (1/6);
        x4(i + 1) = x4(i) + (dx4_k1 + 2 * dx4_k2 + 2 * dx4_k3 + dx4_k4) * dt * (1/6);


    end

end
```



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Contents

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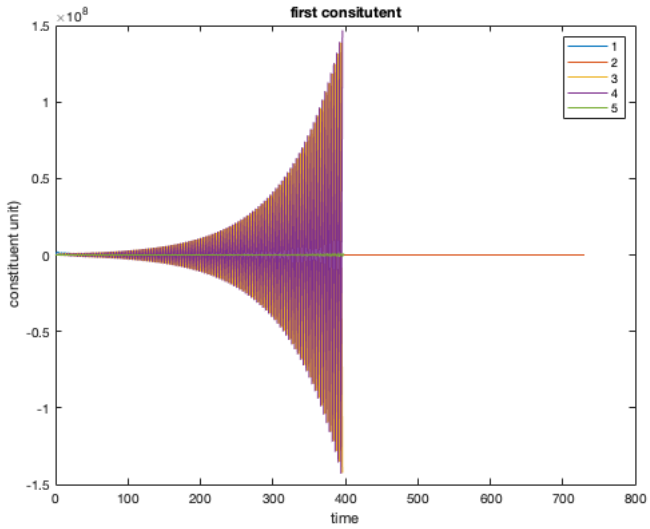
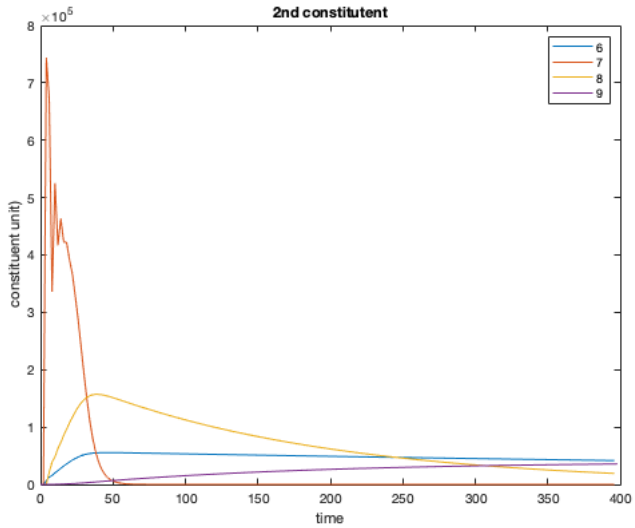
```
%part 2
% a
fprintf("K_mech, which is a function of f_sigma and f_tauw  %f\n")
```

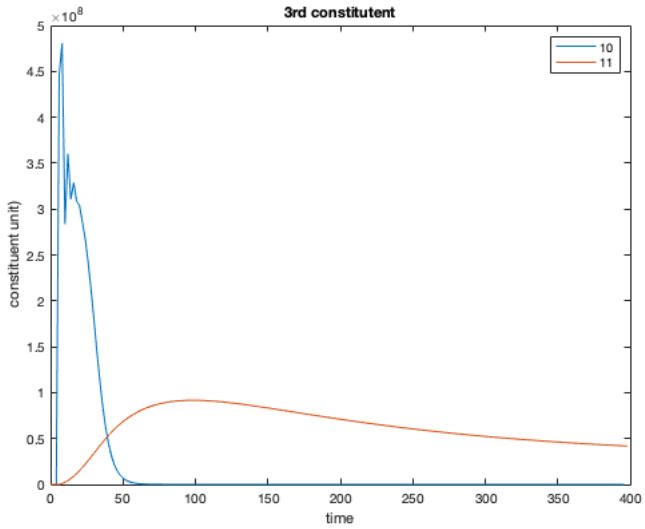
```
K_mech, which is a function of f_sigma and f_tauw
```

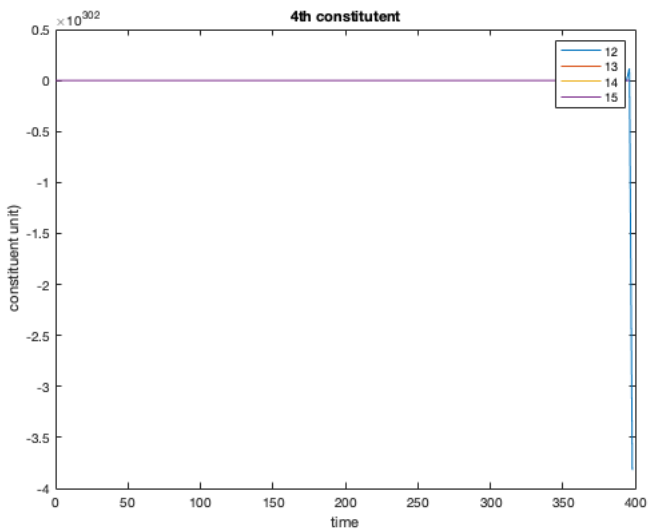
b

```
h=2;
t = 0:h:730 ;
y = [1 0 0 0 0 0 0 0 0 0 0 0 0 0 0]';

prm = write_parameters();

for i = 1: length(t)-1
    step = odefun(t(i),y(:,i),prm);
    y(:,i+1) = y(:,i) + h * step;

end

figure(1)
for p = 1:5;
    plot(t, y(p,:));
    hold on
end
xlabel('time')
ylabel('constituent unit)')
title('first consitutent')
legend('1','2','3','4','5')

figure(2)
for p = 6:9
    plot(t, y(p,:))
    hold on
end
xlabel('time')
ylabel('constituent unit)')
title('2nd constitutent')
legend('6','7','8','9')

figure(3)
for p = 10:11;
    plot(t, y(p,:))
    hold on
end
xlabel('time')
ylabel('constituent unit)')
title('3rd constitutent')
legend('10','11')


figure(4)
for p = 12:15
    plot(t, y(p,:))
    hold on
end
xlabel('time')
ylabel('constituent unit)')
title('4th constitutent')
legend('12','13','14','15')
```


first consitutent
×10^8
1.5
1
0.5
0
-0.5
-1
-1.5
constituent unit)
0
100
200
300
400
500
600
700
800
time
1
2
3
4
5


2nd constitutent
×10^5
8
7
6
5
4
3
2
1
0
constituent unit)
0
50
100
150
200
250
300
350
400
time
6
7
8
9

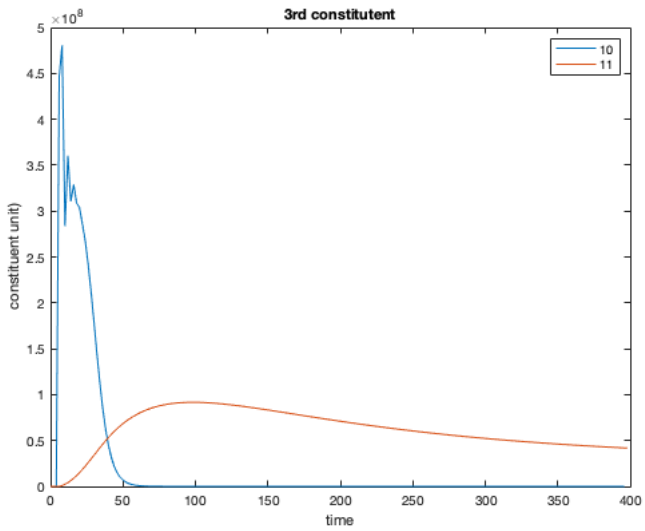


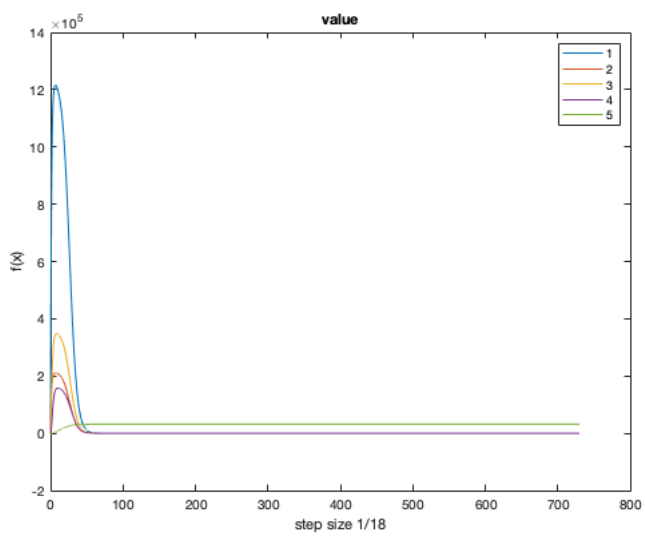


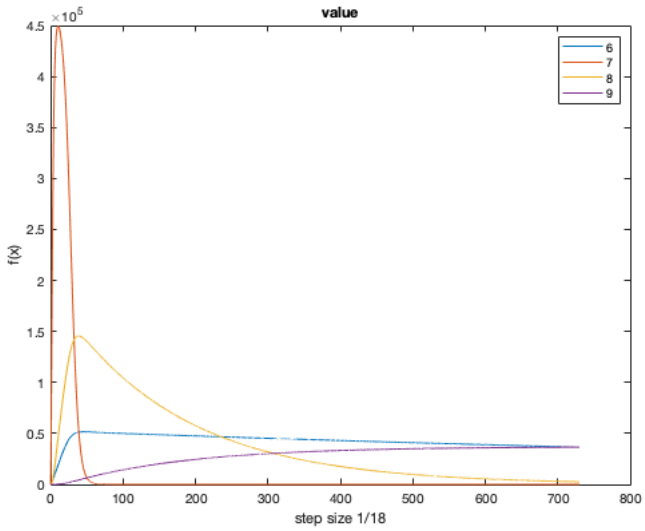
2c

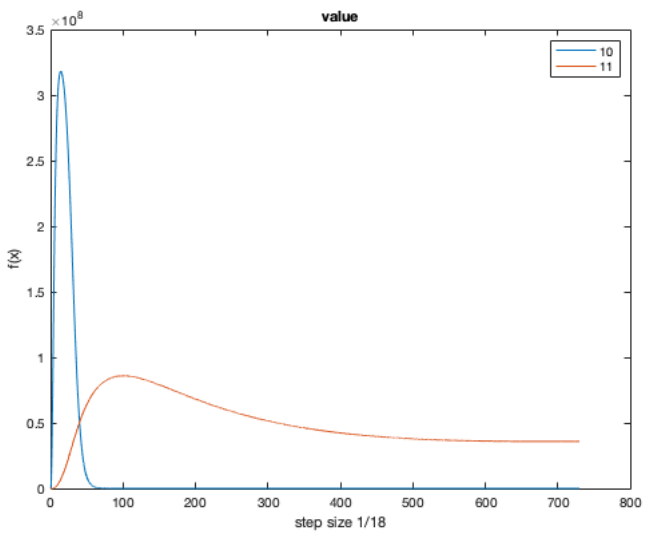
```
h1 = 1/18;

t = 0:h1:730 ;
y = [1 0 0 0 0 0 0 0 0 0 0 0 0 0]';

prm = write_parameters();

for i = 1: length(t)-1
    step = odefun(t(i),y(:,i),prm);
    y(:,i+1) = y(:,i) + h1 * step;

end

figure(5)
for p = 1:5
    plot(t, y(p,:))
    hold on
end
xlabel('step size 1/18')
ylabel('f(x)')
title('value')
legend('1','2','3','4','5')

figure(6)
for p = 6:9;
    plot(t, y(p,:))
    hold on
end
xlabel('step size 1/18')
ylabel('f(x)')
title('value')
legend('6','7','8','9')
```

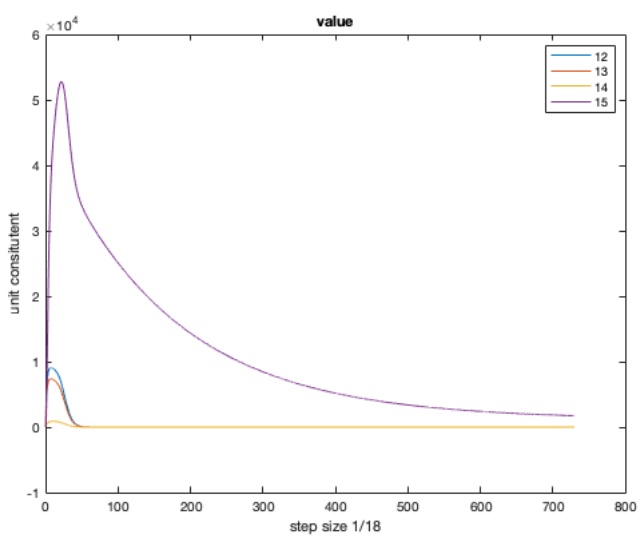
```
figure(7)
for p = 10:11;
    plot(t, y(p,:))
    hold on
end
xlabel('step size 1/18')
ylabel('f(x)')
title('value')
legend('10','11')


figure(8)
for p = 12:15;
    plot(t, y(p,:))
    hold on
end
xlabel('step size 1/18')
ylabel('unit consitutent')
title('value')
legend('12','13','14','15')

% the last constituent requires the smallest time step to stabilize, and
% the first one requries the least time step
```





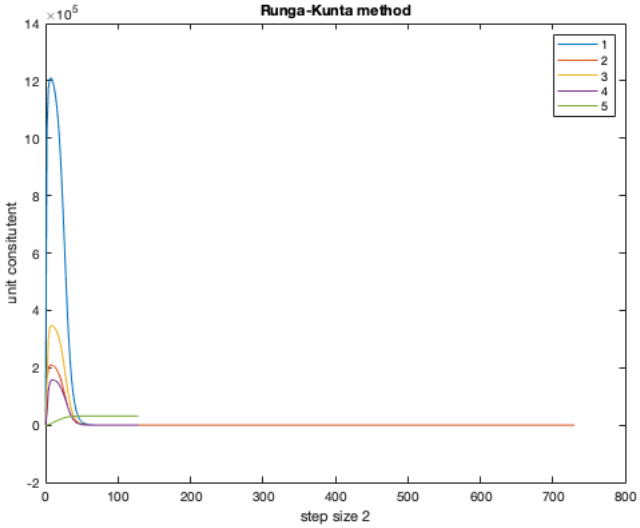


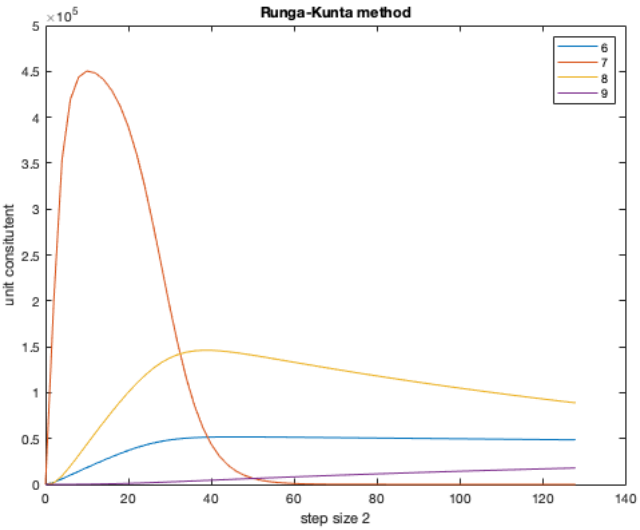


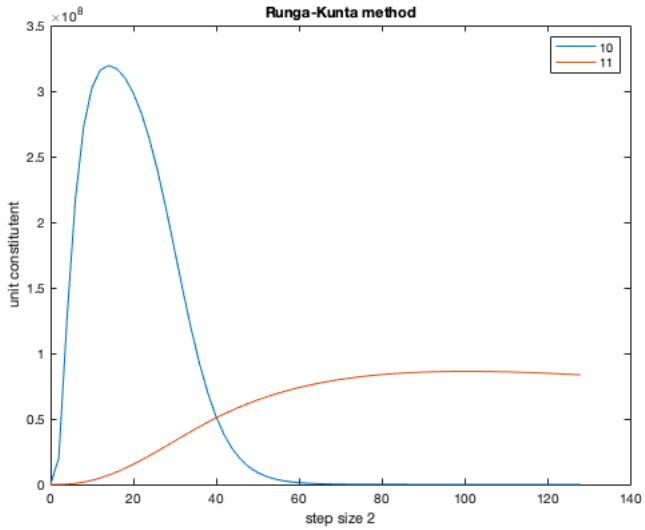


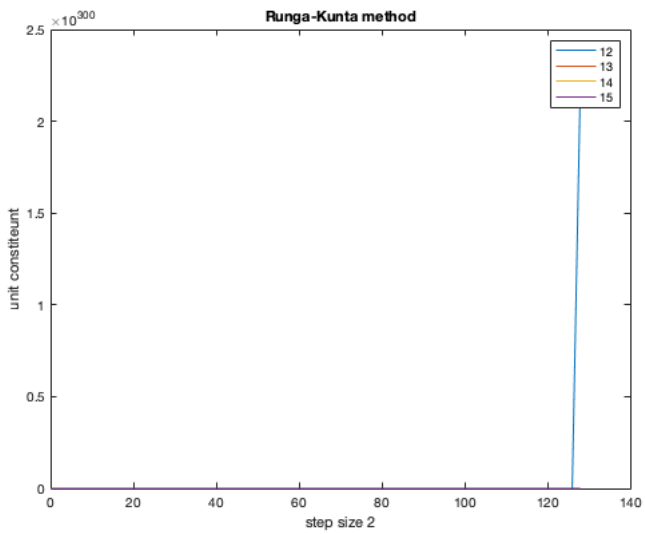

```
%2d i

h = 2 ;

prm = write_parameters();
t = 0:h:730;
y = [1 0 0 0 0 0 0 0 0 0 0 0 0 0 0]';
```

```
value = zeros(length(t),15);
value(1,:) = y;

for i = 1:length(t) - 1

    k1 = h * odefun(t(i), value(i,:),prm);
    k2 = h * odefun(t(i) + h / 2, value(i,:)' + k1 / 2,prm);
    k3 = h * odefun(t(i) + h / 2, value(i,:)' + k2 / 2,prm);
    k4 = h * odefun(t(i) + h, value(i,:)' + k3,prm);
    value(i + 1,:) = value(i,:)' + (k1 + 2 * k2 + 2 * k3 + k4) / 6;
end

figure(9)
for p = 1:5
    plot(t, value(:,p))
    hold on
end
xlabel('step size 2')
ylabel('unit consitutent')
title('Runga-Kunta method')
legend('1','2','3','4','5')

figure(10)
for p = 6:9;
    plot(t, value(:,p))
    hold on
end
xlabel('step size 2')
ylabel('unit consitutent')
title('Runga-Kunta method')
legend('6','7','8','9')


figure(11)
for p = 10:11;
    plot(t, value(:,p))
    hold on
end
xlabel('step size 2')
ylabel('unit constitutent')
title('Runga-Kunta method')
legend('10','11')


figure(12)
for p = 12:15;
    plot(t, value(:,p))
    hold on
end
xlabel('step size 2')
ylabel('unit constiteunt')
title('Runga-Kunta method')
legend('12','13','14','15')


% the last constituent requires the smallest time step to stabilize, and
% the first one requries the least time step
```


Runga-Kunta method
×10^5
14
12
10
8
6
4
2
0
-2
0
100
200
300
400
500
600
700
800
unit consitutent
step size 2
1
2
3
4
5


Runga-Kunta method
×10^5
5
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0
0
20
40
60
80
100
120
140
unit consitutent
step size 2
6
7
8
9





2d ii

```
h3 = 1/18 ;

prm = write_parameters();
t = 0:h3:730;
y = [1 0 0 0 0 0 0 0 0 0 0 0 0 0 0]';
value = zeros(length(t),15);
value(1,:) = y;

for i = 1:length(t) - 1

    k1 = h3 * odefun(t(i), value(i,:),prm);
    k2 = h3 * odefun(t(i) + h3 / 2, value(i,:)' + k1 / 2,prm);
    k3 = h3 * odefun(t(i) + h3 / 2, value(i,:)' + k2 / 2,prm);
    k4 = h3 * odefun(t(i) + h3, value(i,:)' + k3,prm);
    value(i + 1,:) = value(i,:)' + (k1 + 2 * k2 + 2 * k3 + k4) / 6;
end

figure(13)
for p = 1:5
    plot(t, value(:,p))
    hold on
end
xlabel('step size 1/18')
ylabel('unit consitutent')
title('Runga-Kunta method')
legend('1','2','3','4','5')

figure(14)
for p = 6:9;
    plot(t, value(:,p))
    hold on
end
```

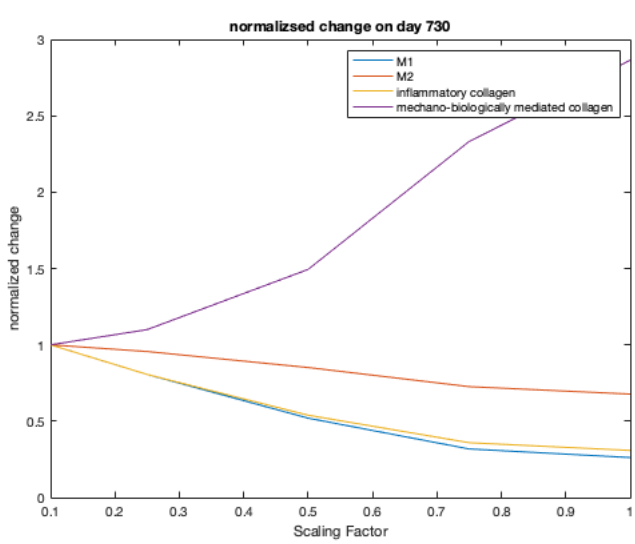
```
xlabel('step size 1/18')
ylabel('unit consitutent')
title('Runga-Kunta method')
legend('6','7','8','9')


figure(15)
for p = 10:11;
    plot(t, value(:,p))
    hold on
end
xlabel('step size 1/18')
ylabel('unit consitutent')
title('Runga-Kunta method')
legend('10','11')


figure(16)
for p = 12:15;
    plot(t, value(:,p))
    hold on
end
xlabel('step size 1/18')
ylabel('unit consitutent')
title('Runga-Kunta method')
legend('12','13','14','15')

% time step needed for euler or RK is similar in order to reach stability
```

```
% 2e i
prm = write_parameters();

h4 = 1/18;

t = 0:h4:730 ;
y = [1 0 0 0 0 0 0 0 0 0 0 0 0 0 0]';
y_matrices = zeros(15, length(t), 5);

for j = 1:5
    temp_param1 = prm.K_TGFB_max;
    scaling_fac = [1,0.75,0.5,0.25,0.1];
    prm.K_TGFB_max = temp_param1*scaling_fac(j);
    for i = 1: length(t)-1;
        step = odefun(t(i),y(:,i),prm);
        y(:,i+1) = y(:,i) + h1 * step;
    end

    y_matrices(:, :, j) = y;
    M1(j,:) = y(3,:)
    M2(j,:) = y(4,:);
    inflamcoll(j,:) = y(10,:);
    mechano_coll(j,:) = y(11,:);
end


result_1 = y_matrices(:, :, 1) ./ y_matrices(:, :, 1);
result_75 = y_matrices(:, :, 2) ./ y_matrices(:, :, 1);
result_50 = y_matrices(:, :, 3) ./ y_matrices(:, :, 1);
result_25 = y_matrices(:, :, 4) ./ y_matrices(:, :, 1);
result_10 = y_matrices(:, :, 5) ./ y_matrices(:, :, 1);


p1 = 3
figure(17)
plot(t, result_1(p1,:))
hold on
plot(t, result_75(p1,:))
plot(t, result_50(p1,:))
plot(t, result_25(p1,:))
plot(t, result_10(p1,:))
xlabel('time')
ylabel('unit constituent')
title('M1 macrophages')
legend('1','0.75','0.5','0.25','0.1')
hold off

p2 = 4
figure(18)
plot(t, result_1(p2,:))
hold on
plot(t, result_75(p2,:))
plot(t, result_50(p2 ,:))
plot(t, result_25(p2,:))
plot(t, result_10(p2,:))
xlabel('time')
ylabel('unit constituent')
```

```
title('M2 macrophages')
legend('1','0.75','0.5','0.25','0.1')
hold off

p4 = 10
figure(19)
plot(t, result_1(p4,:))
hold on
plot(t, result_75(p4,:))
plot(t, result_50(p4,:))
plot(t, result_25(p4,:))
plot(t, result_10(p4,:))
xlabel('time')
ylabel('unit constituent')
title('inflammatory collagen')
legend('1','0.75','0.5','0.25','0.1')
hold off

p4 = 11
figure(20)
plot(t, result_1(p4,:))
hold on
plot(t, result_75(p4,:))
plot(t, result_50(p4,:))
plot(t, result_25(p4,:))
plot(t, result_10(p4,:))
xlabel('time')
ylabel('unit constituent')
title('mechano-biologically mediated collagen')
legend('1','0.75','0.5','0.25','0.1')
hold off
```

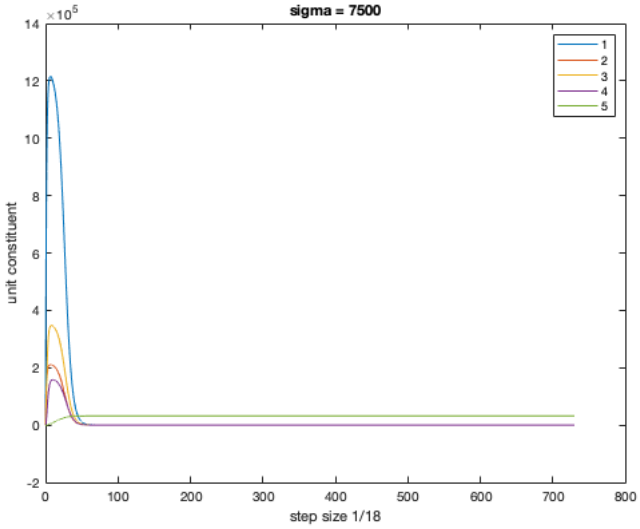
2e ii

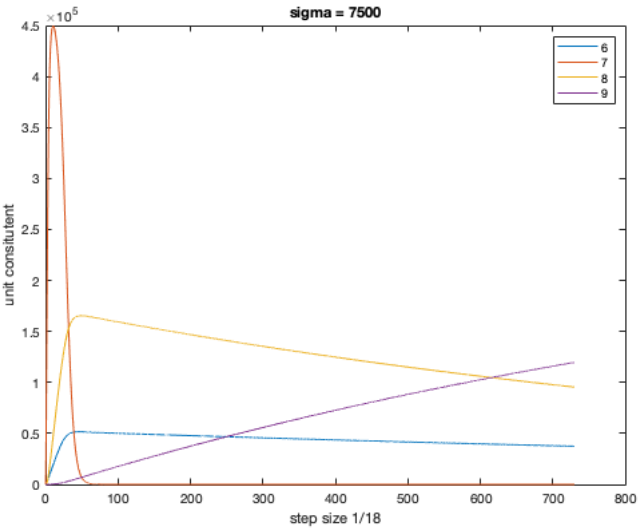
```
y3 = [result_1(3,7301),result_75(3,7301),result_50(3,7301),result_25(3,7301),result_10(3,7301)];
y4 = [result_1(4,7301),result_75(4,7301),result_50(4,7301),result_25(4,7301),result_10(4,7301)];
y10 = [result_1(10,7301),result_75(10,7301),result_50(10,7301),result_25(10,7301),result_10(10,7301)];
y11 = [result_1(11,7301),result_75(11,7301),result_50(11,7301),result_25(11,7301),result_10(11,7301)];

figure(21)
x = [0.1,0.25,0.5,0.75,1];
plot(x,y3)
hold on
plot(x,y4)
hold on
plot(x,y10)
hold on
plot(x,y11)
hold on

xlabel('Scaling Factor')
ylabel('normalized change')
title('normalizsed change on day 730')
legend('M1', 'M2', 'inflammatory collagen', 'mechano-biologically mediated collagen');

% M1, with normalized change of 3, has the largest relative changes with a
% reduction in the effect of TGFB.
```

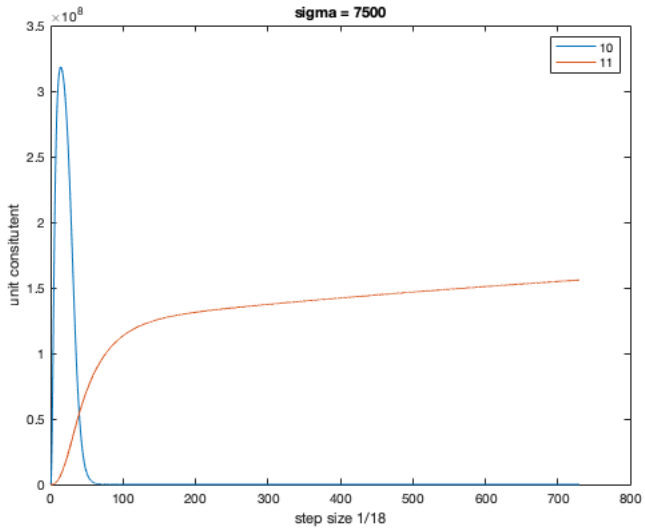


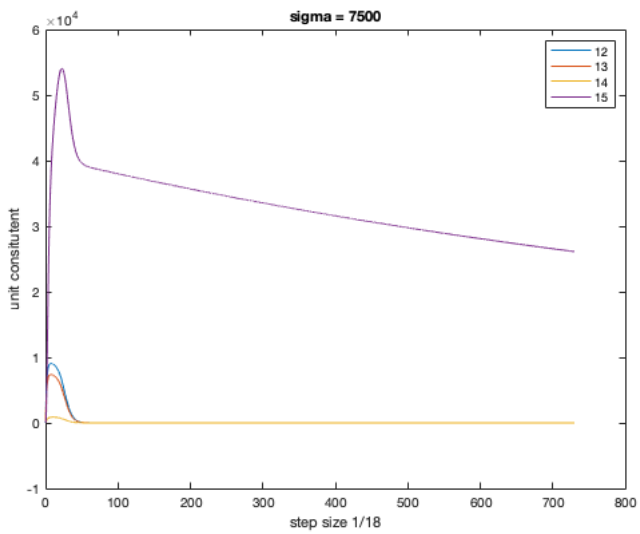
2f i

```
%i sigma is 7500
temp_prm1 = write_parameters();
temp_prm1.sigma = 7500;
temp_prm1.tauw = 2;
h1 = 1/18;

for i = 1: length(t)-1;
    step = odefun(t(i),y(:,i),temp_prm1);
    y(:,i+1) = y(:,i) + h1 * step;
    y1 = y;
end

figure(22)
for p = 1:5
    plot(t, y(p,:))
    hold on
end
xlabel('step size 1/18')
ylabel('unit constituent')
title('sigma = 7500')
legend('1','2','3','4','5')

figure(23)
for p = 6:9;
    plot(t, y(p,:))
    hold on
end
xlabel('step size 1/18')
ylabel('unit consitutent')
title('sigma = 7500')
legend('6','7','8','9')


figure(24)
for p = 10:11;
    plot(t, y(p,:))
    hold on
end
xlabel('step size 1/18')
ylabel('unit consitutent')
title('sigma = 7500')
legend('10','11')


figure(25)
for p = 12:15;
    plot(t, y(p,:))
    hold on
end
xlabel('step size 1/18')
ylabel('unit consitutent')
title('sigma = 7500')
legend('12','13','14','15')


% they all look similar
```


sigma = 7500
×10^5
14
12
10
8
6
4
2
0
-2
0
100
200
300
400
500
600
700
800
unit constituent
step size 1/18
1
2
3
4
5

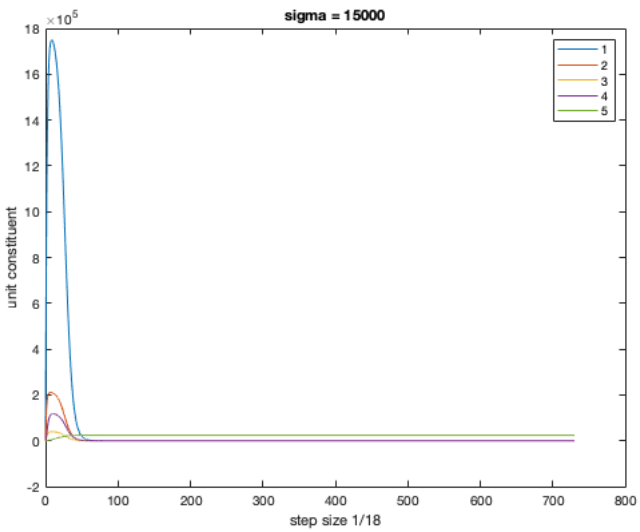

sigma = 7500
×10^5
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0
0
100
200
300
400
500
600
700
800
unit consitutent
step size 1/18
6
7
8
9

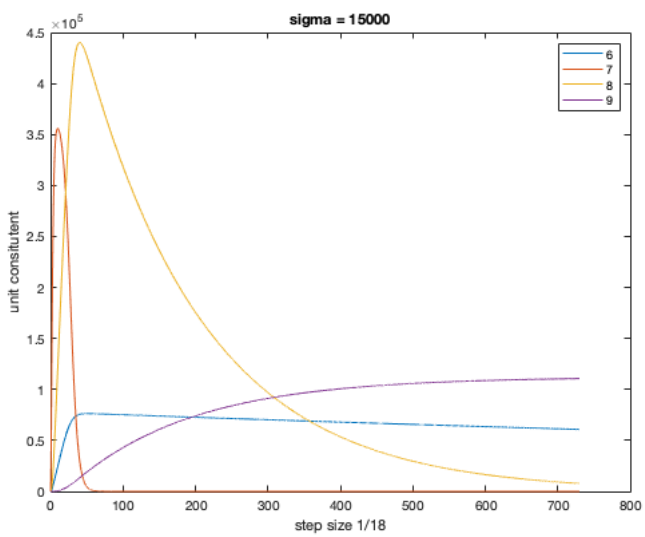


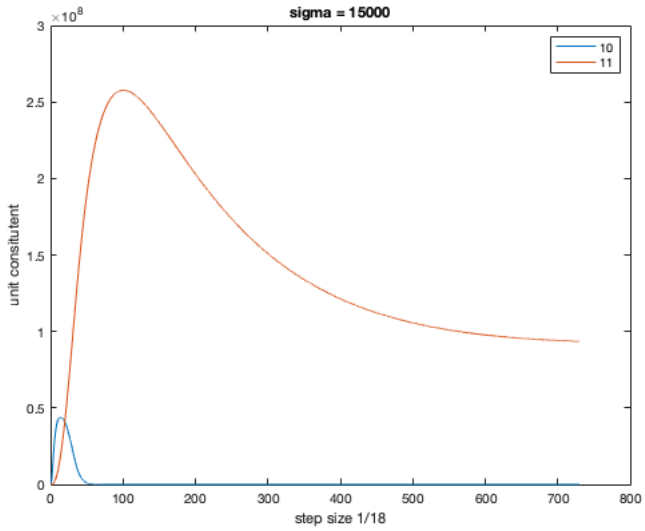


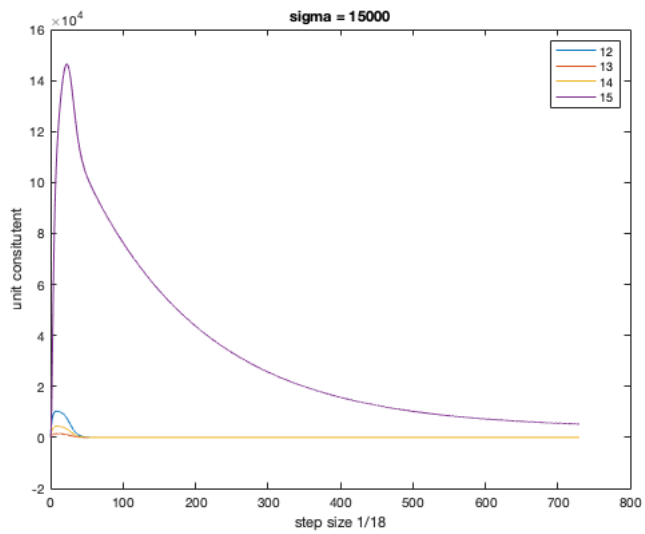

```
% ii sigma is 15000
temp_prm3 = write_parameters();
temp_prm3.sigma = 15000;
temp_prm3.tauw = 2;
h1 = 1/18;
for i = 1: length(t)-1;
    step = odefun(t(i),y(:,i),prm);
    y(:,i+1) = y(:,i) + h1 * step;
    y2 = y ;
end

figure(26)
for p = 1:5
    plot(t, y2(p,:))
    hold on
end
xlabel('step size 1/18')
ylabel('unit constituent')
title('sigma = 15000')
legend('1','2','3','4','5')

figure(27)
for p = 6:9;
    plot(t, y2(p,:))
    hold on
end
xlabel('step size 1/18')
ylabel('unit consitutent')
title('sigma = 15000')
legend('6','7','8','9')


figure(28)
for p = 10:11;
```

```
    plot(t, y2(p,:))
    hold on
end
xlabel('step size 1/18')
ylabel('unit consitutent')
title('sigma = 15000')
legend('10','11')


figure(29)
for p = 12:15;
    plot(t, y2(p,:))
    hold on
end
xlabel('step size 1/18')
ylabel('unit consitutent')
title('sigma = 15000')
legend('12','13','14','15')
```





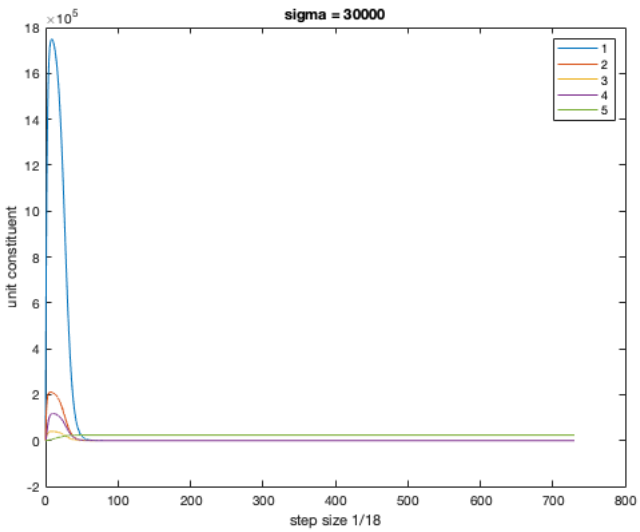


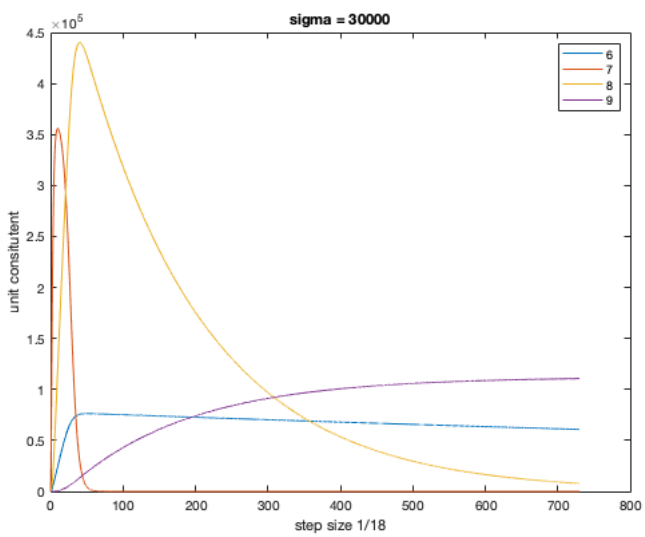


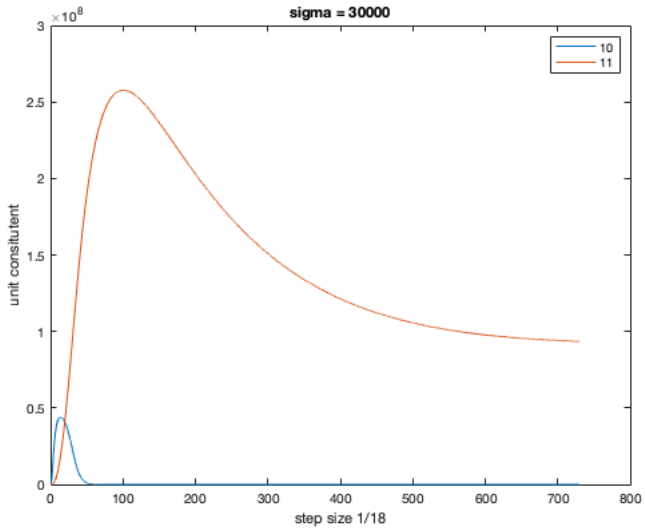
iii sigma is 30000

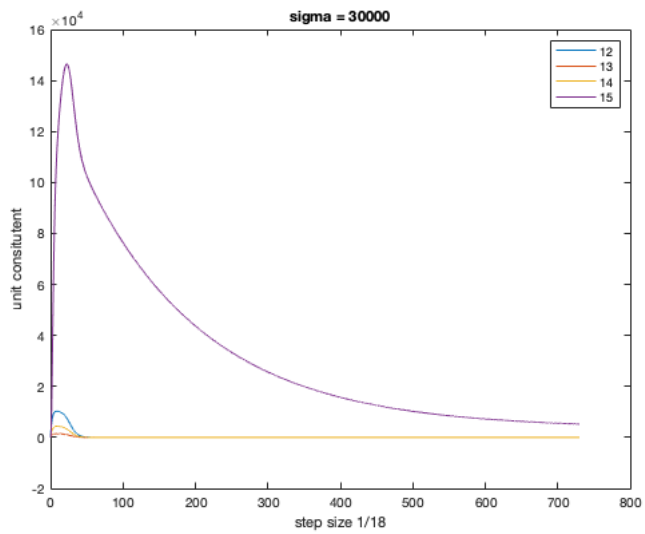
```
temp_prm3 = write_parameters();
temp_prm3.sigma = 30000;
temp_prm3.tauw = 2;
h1 = 1/18;
for i = 1: length(t)-1;
    step = odefun(t(i),y(:,i),prm);
    y(:,i+1) = y(:,i) + h1 * step;
    y3 = y ;
end

figure(30)
for p = 1:5
    plot(t, y3(p,:))
    hold on
end
xlabel('step size 1/18')
ylabel('unit constituent')
title('sigma = 30000')
legend('1','2','3','4','5')

figure(31)
for p = 6:9;
    plot(t, y3(p,:))
    hold on
end
xlabel('step size 1/18')
ylabel('unit consitutent')
title('sigma = 30000')
legend('6','7','8','9')


figure(32)
```

```
for p = 10:11;
    plot(t, y3(p,:))
    hold on
end
xlabel('step size 1/18')
ylabel('unit consitutent')
title('sigma = 30000')
legend('10','11')


figure(33)
for p = 12:15;
    plot(t, y3(p,:))
    hold on
end
xlabel('step size 1/18')
ylabel('unit consitutent')
title('sigma = 30000')
legend('12','13','14','15')
```










```
% iv
% indeed I tried much smaller time step, which doesnt affect the result. at this point, it's more useful to add a higher dimension term, i.e. degradat
```

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